

Foundations of Quantum Computing

Week 3: Single-qubit unitary transformations and multi-qubit states and transformations

Explore – Study

This week we will study measurement and unitaries of multi-qubit states. In this week's Study section you will cover the following topics:

3.1 Single-qubit unitary transformations

3.2 Multi-qubit states and transformations

3.3 Measurement and multi-qubit states

3.1 Single-qubit unitary transformations

In its most basic form, quantum computation is about changing the states of qubits to transform the information they contain, in order to realise a computation. The simplest way that a quantum state can change is by a unitary transformation.

We encountered unitary transformations last week, when discussing ‘normal’ matrices. When we did, we emphasised the properties of their eigenvectors and eigenvalues. This fact is important to the way that certain quantum algorithms work. Also, on a physical level, it is connected to the reason why unitary transformations are relevant to the way that quantum states change. Additionally, the other property of unitary matrices that we encountered, that $U^\dagger U = I$, is the simpler one to explain: it means that for any quantum state-vector $|\psi\rangle$, the vector $U|\psi\rangle$ will also be a quantum state-vector.

We first consider unitary transformations acting on a single qubit, emphasising in particular how you can describe any unitary transformation $U : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ as the result of performing a sequence of simpler transformations.

3.1.1 Preamble: Rotations in 3D space

The way that unitary transformations act on a single qubit is again closely connected with the ‘unit sphere’ in 3D. Because every single-qubit state corresponds to a point on the unit sphere, transformations on single-qubit states correspond to transformations of the unit sphere – which will be by 3D rotations. We now consider how to describe these, as a preliminary to describing single-qubit unitary transformations.

There are a few different ways to describe a rotation in 3D space, but one of the more helpful ones is to describe an **axis of rotation** and an **angle of rotation**.

- When you roll or spin a ball, there is an ‘axis’ about which it rolls or spins. The two points where the ball meets that axis will spin around but stay in place, like the centre of a turntable or carousel. The other points revolve around the line (the ‘axis’) that connects those two points. As with projections and rotations in 2D, an axis of rotation is, in effect, just a direction in space, and can be described by a point on the unit sphere, using angular co-ordinates (θ, ϕ) as we would any other time we wish to describe a point on the sphere.
- For a rotation by a fixed amount, there is also a fixed angle α describing how much the ball turned about the axis. The angle of rotation describes an anticlockwise motion about the axis (for positive angles α) or a clockwise motion about the axis (for negative angles α) – as seen looking at the point on the sphere which we use to describe the rotation axis. (If you point the thumb of your right hand in the direction of the axis of rotation, then the way the fingers of your right hand curl is counterclockwise about that axis.)

A rotation in 3D space is then described by the rotation axis (given by a polar angle θ and azimuthal angle ϕ), and the actual angle α of the rotation.

For a point $\mathbf{v}$ on the unit sphere, we might write $R_{\mathbf{v}}(\alpha)$ for a rotation by an angle α about $\mathbf{v}$. If the rotation is defined about one of the x -, y - or z -axes, then we write the name of an axis in place of $\mathbf{v}$: that is, either $R_x(\alpha)$, $R_y(\alpha)$ or $R_z(\alpha)$. (Figure 3.1 illustrates rotations of the form $R_x(\alpha)$ and $R_z(\alpha)$).

The reason why we consider rotations about the origin in 3D space, is precisely because they preserve the unit sphere: they map every point on the unit sphere, to some other point on the sphere. That also means that if we compose two rotations, one rotation $R_{\mathbf{v}}(\alpha_1)$ followed by another $R_{\mathbf{w}}(\alpha_2)$, the result is another rotation, $R_{\mathbf{w}}(\alpha_2) \cdot R_{\mathbf{v}}(\alpha_1)$. We can describe some simple ways in which these rotation operations act, algebraically:

- A rotation by an angle of 0 about any axis is just the identity map, which is also true for a rotation by any integer multiple of 2π :

$$R_{\mathbf{v}}(\pm 2\pi) = R_{\mathbf{v}}(0) = \text{id}. \quad (3.1)$$

- For two rotations $R_{\mathbf{v}}(\alpha_1)$ and $R_{\mathbf{w}}(\alpha_2)$, there is some other axis $\mathbf{q}$ and some angle β . These describe the composite rotation, $R_{\mathbf{q}}(\beta) = R_{\mathbf{w}}(\alpha_2) \cdot R_{\mathbf{v}}(\alpha_1)$. The axis $\mathbf{q}$ might not have a simple relationship to the axes of rotation $\mathbf{v}$ and $\mathbf{w}$, and the angle β might not have a simple relationship to α_1 and α_2 . But if $\mathbf{v} = \mathbf{w}$ represents the same axis of rotation, then we will have $\mathbf{q} = \mathbf{v}$ as well, and $\beta = \alpha_1 + \alpha_2$:

$$R_{\mathbf{v}}(\alpha_2) \cdot R_{\mathbf{v}}(\alpha_1) = R_{\mathbf{v}}(\alpha_2 + \alpha_1). \quad (3.2)$$

Figure 3.1: Rotation about the x -axis and the z -axis. There are two illustrations of the unit sphere, in terms of rings centred about the x -axis and the z -axis respectively, together with an indication of rotations about these axes.

(a) Illustration of the sphere subject to an anticlockwise rotation about the x -axis. Each ring represents points at some fixed x -coordinate: points on each ring will stay on that ring after a X -rotation.

(b) Illustration of the sphere subject to an anticlockwise rotation about the z -axis. Each ring represents points at some fixed z -coordinate: points on each ring will stay on that ring after a Z -rotation.

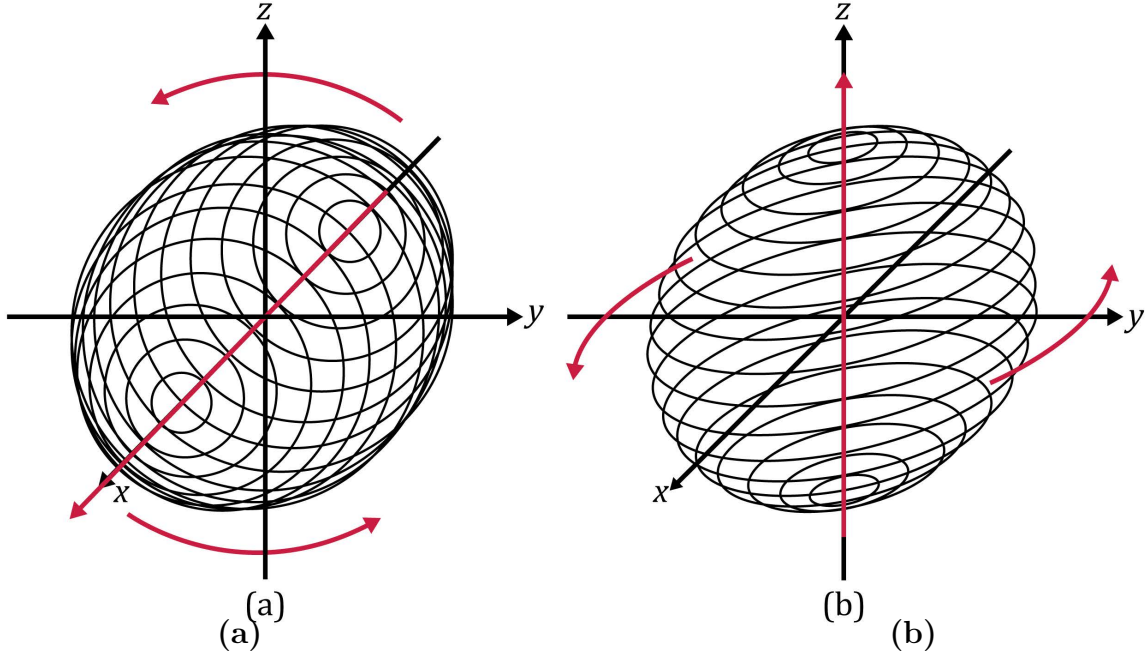


Figure 3.1 Long description: two diagrams in cavalier projection. In each, the x -, y - and z -axes are drawn in grey, and a sphere is drawn in black by means of drawing a collection of oblique ellipses. In the diagram (a), a light red arrow points to the lower-left through the sphere, indicating the positive x direction. The ellipses are centred on this arrow, suggesting circles centred on this arrow as a common axis. There are two curved dark red arrows drawn above and below the sphere, indicating the anticlockwise direction of motion about this axis. In the diagram (b), a light red arrow points straight up, through the sphere, indicating the positive z direction. There are two curved dark red arrows drawn either side of the sphere, indicating the anticlockwise direction of motion about this axis.

- The **inverse** of the rotation $R_v(\alpha)$, is $R_v(-\alpha)$. If we consider the inverse of a rotation $R_q(\beta) = R_w(\alpha_2) \cdot R_v(\alpha_1)$, we can do this by considering how we would **undo** the rotations about v and w , which we would do in the reverse order:

$$R_q(\beta)^{-1} = R_v(\alpha_1)^{-1} \cdot R_w(\alpha_2)^{-1} = R_v(-\alpha_1) \cdot R_w(-\alpha_2). \quad (3.3)$$

These simple algebraic rules will be useful to us, but in practice we will need to describe rotations in more detail to describe how they compose, and how to compute their inverses.

In fact, rotations can be described using linear transformations of 3D space, just as rotations and reflections of the plane can be described by linear transformations of $\mathbb{R}^2$, but because we are more interested in describing transformations of single-qubit state-vectors, we will prefer to describe them in terms of transformations of $\mathbb{C}^2$.

3.1.2 Single-qubit unitary matrices, with examples

Just as 3D rotations are transformations that map the unit sphere to itself, the importance of unitary transformations is that they preserve the set of state-vectors. As we saw last week, unitary transformations U are linear transformations, and preserving the set of state-vectors means in particular that U maps unit vectors to unit vectors. Specifically, for unitary transformations on a single qubit, this means that we are considering a transformation $U : \mathbb{C}^2 \rightarrow \mathbb{C}^2$, such that:

$$\|U|\psi\rangle\| = 1 \quad \text{for any unit vector } |\psi\rangle \in \mathbb{C}^2. \quad (3.4)$$

As $\|U|\psi\rangle\|^2 = (U|\psi\rangle)^\dagger (U|\psi\rangle)$, this means that:

$$\langle\psi|U^\dagger U|\psi\rangle = 1 \quad \text{for any unit vector } |\psi\rangle \in \mathbb{C}^2; \quad (3.5)$$

and from this, it is possible to show that in fact $U^\dagger U = I$. It is this property that many people use to **define** what it means for a matrix to be unitary. For us it serves to justify why unitary transformations are a relevant transformation to perform to a state-vector.

One of the ways in which unitary transformations are important, is as a description of a change in the **way that information is represented**. The way that it does this is by mapping one orthonormal basis to another. Let $|u_0\rangle, |u_1\rangle \in \mathbb{C}^2$ be any orthonormal basis: any pair of single-qubit states, representing opposite directions on the Bloch sphere, so that $\langle u_0 | u_1 \rangle = 0$. For a unitary transformation $U : \mathbb{C}^2 \rightarrow \mathbb{C}^2$, let $|v_0\rangle = U|u_0\rangle$ and $|v_1\rangle = U|u_1\rangle$, then:

$$\langle v_0 | v_1 \rangle = (U|u_0\rangle)^\dagger (U|u_1\rangle) = \langle u_0 | U^\dagger U | u_1 \rangle = \langle u_0 | u_1 \rangle = 0. \quad (3.6)$$

So U maps the orthonormal basis $|u_0\rangle, |u_1\rangle \in \mathbb{C}^2$ to a different orthonormal basis $|v_0\rangle, |v_1\rangle \in \mathbb{C}^2$ – preserving the distinguishability between these pairs of states. This is part of why unitary transformations can be represented by rotations of the unit sphere (specifically, of the Bloch sphere). Like rotations, they preserve the information of how different the single-qubit states are.

Because unitary matrices U are ‘normal’ matrices, they have a pair of orthogonal eigenvectors: we have $U = a_0 |u_0\rangle\langle u_0| + a_1 |u_1\rangle\langle u_1|$ for some orthonormal basis $|u_0\rangle, |u_1\rangle \in \mathbb{C}^2$ and some eigenvalues $u_0, u_1 \in \mathbb{C}$. The state-vectors $|u_0\rangle$ and $|u_1\rangle$ are then unaffected by U , except for a possible global phase factor if $a_0 \neq 1$ or $a_1 \neq 1$, which in any case does not make a difference to what state is being represented. This means that the orthogonal vectors $|u_0\rangle, |u_1\rangle$ represent the two opposite points on the Bloch sphere, which lie on

Figure 3.2: The X, Y and Z bases of qubits marked on the Bloch sphere. The single-qubit states lying on the x -axis (that is $|+\rangle$ and $|-\rangle$), the y -axis (that is $|i\rangle$ and $|-i\rangle$) and the z -axis (that is $|0\rangle$ and $|1\rangle$) of the Bloch sphere. Because of their connection to the operators X , Y and Z described below, we sometimes refer to these as the X -eigenbasis, the Y -eigenbasis and the Z -eigenbasis.

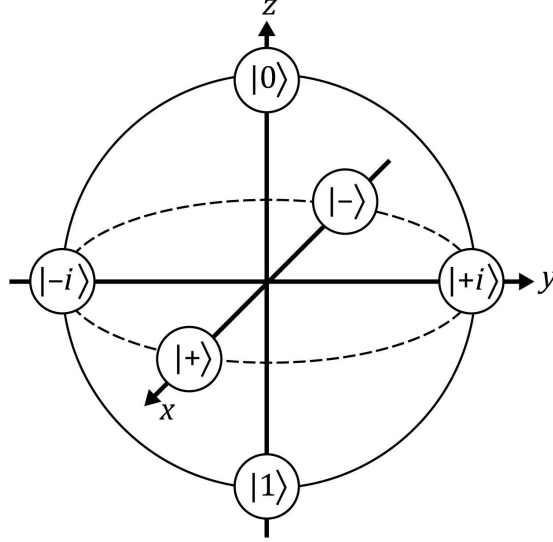


Figure 3.2 Long description: Three coordinate axes in cavalier projection are marked in black. There is a circle drawn to indicate the outline of a sphere, with a dashed ellipse also drawn to help indicate depth. The positive directions of the axes are marked with an appropriate letter and a small arrow. The six points where these axes intersect with the sphere are marked. The intersection of the positive z -axis and the sphere (which is at the north pole) is marked $|0\rangle$. The corresponding intersection of the sphere and the negative z -axis is marked $|1\rangle$. The intersections of the sphere and the positive and negative x -axes are marked $|+\rangle$ and $|-\rangle$. The intersections of the sphere and the positive and negative y -axes are marked $|+i\rangle$ and $|-i\rangle$.

the axis of rotation described by U . For rotations about the x -axis, these would be the states:

$$|u_0\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad \text{and} \quad |u_1\rangle = |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle); \quad (3.7)$$

for rotations about the y -axis, these would be:

$$|u_0\rangle = |+i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle) \quad \text{and} \quad |u_1\rangle = |-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle); \quad (3.8)$$

and for rotations about the z -axis, these would simply be $|u_0\rangle = |0\rangle$ and $|u_1\rangle = |1\rangle$. These state-vectors and their corresponding locations on the Bloch sphere are illustrated in Figure 3.2.

Next, we consider some examples of single-qubit unitaries, described also in terms of rotations of the Bloch sphere.

(a) The Hadamard operation

We begin with one of the more prominent and interesting unitary operations: the Hadamard operation H ,

$$H = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}. \quad (3.9)$$

Essentially, its role is to interchange between the standard basis, the states $|0\rangle$ and $|1\rangle$ that lie on the z -axis of the Bloch sphere, and the states $|+\rangle, |-\rangle$ that lie on the x -axis of the Bloch sphere, which we can see from the columns of H itself:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle, \quad H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle. \quad (3.10)$$

As it happens, H is its own inverse, which means that it also maps the $|+\rangle, |-\rangle$ basis to the computational basis:

$$\begin{aligned} H|+\rangle &= \frac{1}{\sqrt{2}}(H|0\rangle + H|1\rangle) = \frac{1}{2}(|0\rangle + |1\rangle + |0\rangle - |1\rangle) = |0\rangle, \\ H|-\rangle &= \frac{1}{\sqrt{2}}(H|0\rangle - H|1\rangle) = \frac{1}{2}(|0\rangle + |1\rangle - |0\rangle + |1\rangle) = |1\rangle. \end{aligned} \quad (3.11)$$

(It is not difficult to show that it also maps $|+i\rangle \xrightarrow{H} \frac{1+i}{\sqrt{2}}| -i\rangle$ and $| -i\rangle \xrightarrow{H} \frac{1-i}{\sqrt{2}}| +i\rangle$, meaning that it also maps the positive y -axis onto the negative y -axis and vice versa).

These properties are the most important aspects of the Hadamard operation. But if you want to visualise it as a rotation of the Bloch sphere (illustrated in Figure 3.3), we can infer the following things about it:

- (i) Notice that $H^2 = I$: that is, H is self-inverse, so that applying it twice results in the identity operation. It then corresponds to a rotation by an angle of π .
- (ii) Because it interchanges the x -axis and the z -axis, we can describe it by a rotation about an axis which is halfway between the two.

Unlike the other operations we consider in the material that follows, the Hadamard operation is a rotation by a specific angle, by an axis which is not even the x -, y - or z -axis: but it plays an important role in the theory of quantum algorithms.

(b) The X operation and x -rotations

A less-exciting, but still very important single-qubit unitary operation, is the X operation:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \quad (3.12)$$

You may recognise this as being the same as the matrix for ‘NOT’ operation on single-bit probability distributions. (For this reason, we frequently also call X the ‘NOT’ operation). While single-qubit states are **not** probability distributions, we are representing

Figure 3.3: The Hadamard rotation, represented as a rotation of the Bloch sphere. We have indicated the sphere as a sequence of rings about an axis, which is tilted halfway between the x - and z -axes. The intersections of the sphere and the positive x - and z -axes are marked in dark red, and the arc those two points trace out during the rotation has also been marked in dark red.

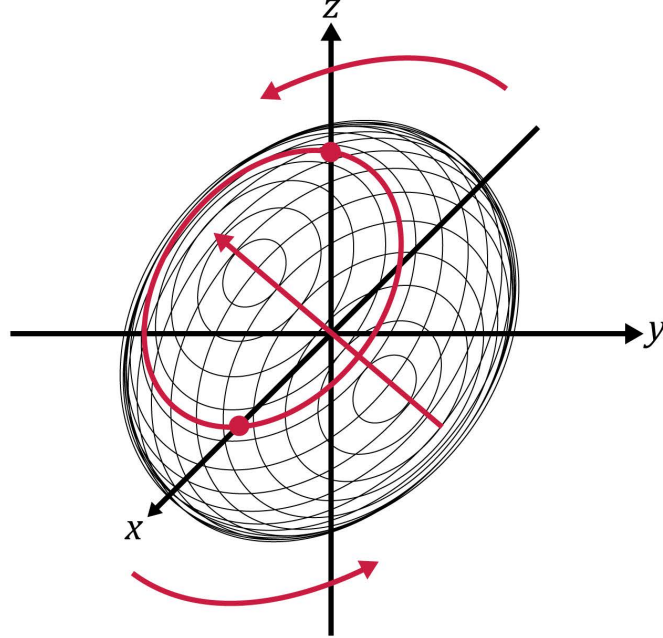


Figure 3.3 Long description: A diagram in cavalier projection. The x -, y - and z -axes are drawn in grey. A sphere is drawn in black by means of drawing a collection of oblique ellipses. A red arrow points up and left, through the sphere, indicating an axis halfway between the x - and z -axes, at a polar angle of $\pi/4$ and an azimuthal angle of 0. There are two curved red arrows drawn either side of the sphere, indicating the anticlockwise direction of motion.

them in a similar way: by two-dimensional vectors $|\psi\rangle = a|0\rangle + b|1\rangle$ representing some sort of combination of ‘definite’ bit-values $|0\rangle$ and $|1\rangle$ – and the X operation has the same effect of interchanging them:

$$X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle. \quad (3.13)$$

However, the X operation leaves the two states $|\pm\rangle$ unchanged (and only changes the state-vector representation of $|-\rangle$ by a global phase factor):

$$\begin{aligned} X|+\rangle &= \frac{1}{\sqrt{2}}(X|0\rangle + X|1\rangle) = \frac{1}{\sqrt{2}}(|1\rangle + |0\rangle) = |+\rangle, \\ X|-\rangle &= \frac{1}{\sqrt{2}}(X|0\rangle - X|1\rangle) = \frac{1}{\sqrt{2}}(|1\rangle - |0\rangle) = -|-\rangle. \end{aligned} \quad (3.14)$$

That is, the state-vectors $|+\rangle$ and $|-\rangle$ are eigenvectors of the X operation: we can decompose it as $X = |+\rangle\langle+| - |-\rangle\langle-|$. These states lie on the x -axis of the Bloch sphere

(which is in fact why this operator is called ‘ X ’): then X can be represented as a rotation by π about the x -axis.

We are also primarily interested in rotations about the x -axis, by angles $\alpha \in \mathbb{R}$ other than $\alpha = \pi$. You can think of these as ‘fractions’ of a NOT operation: something which is impossible or ‘non-sense’ for classical computations, but which is possible because of the continuous state-space that a qubit has. (As a rotation of the Bloch sphere, these would be represented just as in Figure 3.1(a)).

The way in which these rotations would be most often represented **in physics**, is as an operator of the form:

$$\cos(\alpha/2)I - i \sin(\alpha/2)X = \begin{bmatrix} \cos(\frac{\alpha}{2}) & -i \sin(\frac{\alpha}{2}) \\ -i \sin(\frac{\alpha}{2}) & \cos(\frac{\alpha}{2}) \end{bmatrix}. \quad (3.15)$$

There are good reasons for preferring such a representation in some cases. However, in this module, we prefer to represent such a rotation by the matrix:

$$R_x(\alpha) = |+\rangle\langle+| + e^{i\alpha} |-\rangle\langle-| = \frac{1}{2} \begin{bmatrix} 1 + e^{i\alpha} & 1 - e^{i\alpha} \\ 1 - e^{i\alpha} & 1 + e^{i\alpha} \end{bmatrix}. \quad (3.16)$$

The reason why we have two different representations for the same operation, is that these two representations are different only by a global phase factor; specifically:

$$\tilde{R}_x(\alpha) = \cos(\alpha/2)I - i \sin(\alpha/2)X = e^{-i\alpha/2} |+\rangle\langle+| + e^{i\alpha/2} |-\rangle\langle-| = e^{-i\alpha/2} R_x(\alpha). \quad (3.17)$$

The difference between ‘ $\tilde{R}_x(\alpha)$ ’ and $R_x(\alpha)$ is just the factor of $e^{-i\alpha/2}$, so that for any single-qubit state-vector $|\psi\rangle \in \mathbb{C}^2$:

$$|\psi'\rangle = R_x(\alpha) |\psi\rangle \implies \tilde{R}_x(\alpha) |\psi\rangle = e^{-i\alpha/2} |\psi'\rangle, \quad (3.18)$$

where $|\psi'\rangle$ and $e^{-i\alpha/2} |\psi'\rangle$ represent the same single-qubit state. The two matrices $\tilde{R}_x(\alpha)$ and $R_x(\alpha)$ are different, but they represent the same transformation of single-qubit states.

(c) The Z operation and z -rotations

Another important single-qubit operation is the Z operation:

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = |0\rangle\langle 0| - |1\rangle\langle 1|. \quad (3.19)$$

It is easy to see that $Z|0\rangle = |0\rangle$ and $Z|1\rangle = -|1\rangle$. In fact, this operation has no effect on the standard basis states, represented by the state-vectors $|0\rangle$ and $|1\rangle$. What it does have an effect on are the states $|\pm\rangle$ which lie on the x -axis:

$$\begin{aligned} Z|+\rangle &= \frac{1}{\sqrt{2}}(Z|0\rangle + Z|1\rangle) = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle, \\ Z|-\rangle &= \frac{1}{\sqrt{2}}(Z|0\rangle - Z|1\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle. \end{aligned} \quad (3.20)$$

That is, it has the same relationship to the $|\pm\rangle$ basis, as the X operation has with the $|0\rangle, |1\rangle$ basis. The reason for this is in part because Z represents a π rotation about the z -axis. This is something that we can understand from the fact that it is self-inverse and leaves the states on the z -axis unaffected. Then it revolves the state $|+\rangle$ to the position of the state $|-\rangle$, and $|-\rangle$ to the position of the state $|+\rangle$.

The fact that X and Z perform similar operations to the states $|0\rangle, |1\rangle$ and $|+\rangle, |-\rangle$ is something that we can describe using the Hadamard operation. Both X and Z are rotations by π , about the x -axis and the z -axis respectively. That means that they are **similar** operations: we can relate the two by an operation that, for instance, interchanges the x -axis onto the z -axis. Specifically, the X operation could be realised by mapping the x -axis onto the z -axis, performing a Z operation and then undoing the first mapping to move the z -axis back onto the x -axis. Because the Hadamard operation maps the x -axis onto the z -axis (and because H is self-inverse), that means that:

$$X = H^{-1}ZH = HZH. \quad (3.21)$$

We similarly have $Z = HXH$: we can equivalently think of Z as something we could realise by mapping the z -axis onto the x -axis, performing an X operation and then undoing the first operation in order to restore the z -axis to its original orientation.

We are interested in other rotations about the z -axis as well: as rotations of the Bloch sphere, these would be represented just as in Figure 3.1(b). The way in which these rotations would be most often represented in physics, is as an operator of the form:

$$\cos(\alpha/2)I - i\sin(\alpha/2)Z = \begin{bmatrix} \cos(\frac{\alpha}{2}) - i\sin(\frac{\alpha}{2}) & 0 \\ 0 & \cos(\frac{\alpha}{2}) + i\sin(\frac{\alpha}{2}) \end{bmatrix} = \begin{bmatrix} e^{-i\alpha/2} & 0 \\ 0 & e^{i\alpha/2} \end{bmatrix}. \quad (3.22)$$

However, we will prefer to represent z -axis rotations by α by:

$$R_z(\alpha) = |0\rangle\langle 0| + e^{i\alpha}|1\rangle\langle 1| = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\alpha} \end{bmatrix}. \quad (3.23)$$

Again, these operators differ only by a global phase, so represent the same transformation of single-qubit states.

(d) The Y operation and y -axis rotations

You may wonder whether there is also a Y operation, which corresponds to a y -axis rotation by an angle of π . The answer is ‘yes’:

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = |+i\rangle\langle +i| - |-i\rangle\langle -i|. \quad (3.24)$$

It is easy to see that $Y|+i\rangle = |+i\rangle$ and $Y|-i\rangle = -|-i\rangle$. In fact, we can also see that $Y|0\rangle = i|1\rangle$ and $Y|1\rangle = -i|0\rangle$, which means that – apart from undetectable global phases which don’t refer to any differences in the state of the qubit – the Y operation has

the same effect on the standard basis states $|0\rangle, |1\rangle$ as the X operation does. Similarly, we can show that:

$$\begin{aligned} Y|+\rangle &= \frac{1}{\sqrt{2}}(Y|0\rangle + Y|1\rangle) = \frac{1}{\sqrt{2}}(i|1\rangle - i|0\rangle) = -i|-\rangle, \\ Y|-\rangle &= \frac{1}{\sqrt{2}}(Y|0\rangle - Y|1\rangle) = \frac{1}{\sqrt{2}}(i|1\rangle + i|0\rangle) = i|+\rangle. \end{aligned} \quad (3.25)$$

Then the Y operation also interchanges the $|\pm\rangle$ states, as again the global phase factors of i and $-i$ do not refer to any property of the states themselves.

As with x - and z -axis rotations, we are interested in other rotations about the y -axis as well. The way in which these rotations would be most often represented in physics, is in this case certainly a convenient one:

$$\tilde{R}_y(\alpha) = \cos(\alpha/2)I - i\sin(\alpha/2)Y = \begin{bmatrix} \cos(\frac{\alpha}{2}) & -\sin(\frac{\alpha}{2}) \\ \sin(\frac{\alpha}{2}) & \cos(\frac{\alpha}{2}) \end{bmatrix}, \quad (3.26)$$

which is exactly the same as a rotation by $\alpha/2$ in the 2D plane. This is related to the role that the polar angle θ plays in describing single-qubit states, when we consider a state corresponding to the angular co-ordinates (θ, ϕ) on the Bloch sphere. In fact, we can read directly from the first column of the matrix above, that it maps $|0\rangle \mapsto \cos(\frac{\alpha}{2})|0\rangle + \sin(\frac{\alpha}{2})|1\rangle$. In the specific case of $\alpha = \pi/2$, it also provides us with an **alternative** operation to interchange the x - and z -axes, albeit mapping the positive x -axis onto the negative z -axis:

$$\tilde{R}_y(\pi/2)|0\rangle = \cos(\frac{\pi}{4})|0\rangle + \sin(\frac{\pi}{4})|1\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle = |+\rangle; \quad (3.27)$$

$$\tilde{R}_y(\pi/2)|+\rangle = \frac{1}{\sqrt{2}}(\tilde{R}_y(\pi/2)|0\rangle + \tilde{R}_y(\pi/2)|1\rangle) = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle) = |1\rangle; \quad (3.28)$$

$$\tilde{R}_y(\pi/2)|1\rangle = -\sin(\frac{\pi}{4})|0\rangle + \cos(\frac{\pi}{4})|1\rangle = -\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle = -|-\rangle; \quad (3.29)$$

$$\tilde{R}_y(\pi/2)|-\rangle = \frac{1}{\sqrt{2}}(\tilde{R}_y(\pi/2)|0\rangle - \tilde{R}_y(\pi/2)|1\rangle) = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle) = |0\rangle. \quad (3.30)$$

Notice, however, the factor of -1 introduced in Equation (3.29). This makes no difference to the single-qubit state. However, it does mean that when we consider repeated applications of $\tilde{R}_y(\pi/2)$, we have:

$$|0\rangle \xrightarrow{\tilde{R}_y(\pi/2)} |+\rangle \xrightarrow{\tilde{R}_y(\pi/2)} |1\rangle \xrightarrow{\tilde{R}_y(\pi/2)} -|-\rangle \xrightarrow{\tilde{R}_y(\pi/2)} -|0\rangle, \quad (3.31)$$

which represents the same state as $|0\rangle$, but with a superfluous factor of -1 . However nice it is to represent y -axis rotations by a 2×2 real-valued matrix, if we want to have a representation in which four rotations by $\pi/2$ represent a full rotation with no effect on state-vectors, then it will be better to use the representation:

$$R_y(\alpha) = |+\rangle\langle +| + e^{i\alpha}|-\rangle\langle -| = e^{i\alpha/2} \begin{bmatrix} \cos(\frac{\alpha}{2}) & -\sin(\frac{\alpha}{2}) \\ \sin(\frac{\alpha}{2}) & \cos(\frac{\alpha}{2}) \end{bmatrix}. \quad (3.32)$$

Again, these operators differ only by a global phase, so represent the same transformation of single-qubit states.

3.1.3 The Pauli operators

The operators X , Y and Z are important 2×2 complex matrices. They each have the special property of unitary and self-inverse, which means they are all also Hermitian (because their eigenvalues are each ± 1). They are also related to one another, through the equations:

$$XY = iZ, \quad YZ = iX, \quad ZX = iY. \quad (3.33)$$

These matrices also **anti-commute**, which is a fancy way of saying that you can change the order of multiplication if you also introduce a factor of -1 :

$$XY = -YX, \quad YZ = -ZY, \quad ZX = -XZ. \quad (3.34)$$

These operators play many important roles in quantum mechanics, owing in part to the fact that any 2×2 matrix can be described as a linear combination of I , X , Y and Z : for an arbitrary 2×2 matrix M , we have:

$$M = \begin{bmatrix} m_{00} & m_{01} \\ m_{10} & m_{11} \end{bmatrix} = \left(\frac{m_{00}+m_{11}}{2}\right)I + \left(\frac{m_{01}+m_{10}}{2}\right)X + \left(\frac{im_{01}-im_{10}}{2}\right)Y + \left(\frac{m_{00}-m_{11}}{2}\right)Z. \quad (3.35)$$

In particular, as the Pauli matrices X , Y and Z anti-commute, it is possible to decompose 2×2 unitary matrices as $U = aI + bX + cY + dZ$, where $|a|^2 + |b|^2 + |c|^2 + |d|^2 = 1$. There are more interesting properties of this sort that the Pauli operators satisfy. For now, we are content to take note of the importance of these operators.

3.1.4 Decomposing single-qubit unitaries

Just as it is practical to describe complicated Boolean functions $f : \{0, 1\}^m \rightarrow \{0, 1\}^n$ in terms of the simpler logical operators ‘NOT’, ‘AND’, ‘OR’ and ‘XOR’, it will be practical for us to describe arbitrary rotations of the sphere in terms of x -, y - and z -axis rotations. We describe how this may be done next.

(a) Decomposing single-qubit unitaries in terms of their axis of rotation

The descriptions of $R_x(\alpha)$, $R_y(\alpha)$ and $R_z(\alpha)$ all follow a pattern: they have a $+1$ -eigenstate (a state which is represented by an eigenvector) corresponding to the state in the direction of the axis of rotation and an $e^{i\alpha}$ -eigenstate consisting of an orthogonal unit vector. We can do something similar for any single-qubit transformation. Because any single-qubit unitary transformation $U : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ has an eigen-decomposition $U = a_0 |u_0\rangle\langle u_0| + a_1 |u_1\rangle\langle u_1|$, where both a_0 and a_1 are complex phases, we can consider an equivalent unitary operation:

$$U' = |u_0\rangle\langle u_0| + a_0^* a_1 |u_1\rangle\langle u_1| \quad (3.36)$$

where $a_0^* a_1$ is itself a complex phase factor. Then $U' = a_0 U$ is a unitary transformation which is equivalent to U , with a $+1$ -eigenvector $|u_0\rangle$ which characterises the axis of rotation of the Bloch sphere that it represents, and $a_0^* a_1 = e^{i\alpha}$ for the rotation angle $\alpha \in \mathbb{R}$ of the Bloch sphere.

By considering how we can relate the $|u_0\rangle, |u_1\rangle$ basis to the $|0\rangle, |1\rangle$ basis, we can describe a way to decompose this transformation in a way that does not depend on the specific value of the rotation angle α . Following the relationship described in Week 2, between single-qubit states and angular co-ordinates of points on the Bloch sphere, suppose that $|u_0\rangle = \cos(\frac{\theta}{2})|0\rangle + e^{i\phi}\sin(\frac{\theta}{2})|1\rangle$ for a polar angle θ and azimuthal angle ϕ . Using the preceding descriptions for y -rotations and z -rotations, we then have:

$$|u_0\rangle = e^{i\theta/2}R_z(\phi)R_y(\theta)|0\rangle. \quad (3.37)$$

Let $Q = R_z(\phi)R_y(\theta)$: then $Q^\dagger|u_0\rangle$ differs from $|0\rangle$ only by a global phase. Because unitary transformations map orthogonal pairs of vectors to other orthogonal pairs of vectors, $Q^\dagger|u_1\rangle$ must be orthogonal to $|0\rangle$. This means that it must differ from $|1\rangle$ by only a global phase. We can then show that:

$$Q^\dagger|u_0\rangle\langle u_0|Q = |0\rangle\langle 0|, \quad Q^\dagger|u_1\rangle\langle u_1|Q = |1\rangle\langle 1|. \quad (3.38)$$

As a result, we have:

$$Q^\dagger U' Q = Q^\dagger \left(|u_0\rangle\langle u_0| + e^{i\alpha} |u_1\rangle\langle u_1| \right) Q = |0\rangle\langle 0| + e^{i\alpha} |1\rangle\langle 1| = R_z(\alpha). \quad (3.39)$$

By isolating the operator U' and expanding the definition of Q , we then obtain:

$$U' = QR_z(\alpha)Q^\dagger = R_z(\phi)R_y(\theta)R_z(\alpha)R_y(-\theta)R_z(-\phi). \quad (3.40)$$

In this way, we can obtain a description of how to realise the transformation U , just in terms of the rotation angle α that it realises on the Bloch sphere, and from the angular co-ordinates of the eigenvector $|u_0\rangle$. Furthermore, to describe any other rotation α' , we may use the same decomposition, except replacing the $R_z(\alpha)$ operation with $R_z(\alpha')$.

(b) Euler angle decompositions

The preceding shows that we can realise the effect of any unitary transformation U of single-qubit states, using just y -axis and z -axis rotations. (We can obtain a similar result using any pair of the x -, y - and z -axes. For instance, to do the same using x -axis and z -axis rotations, we can simply decompose $R_y(\theta)$ as $R_z(\frac{\pi}{2})R_x(\theta)R_z(-\frac{\pi}{2})$ as needed, and combine the z -axis rotations as needed).

However, if we want to describe a single unitary transformation U , rather than a range of transformations which rotate the Bloch sphere by different angles about a common axis, we can reduce the number of rotations involved from five to three, using what are known as **Euler angle decompositions**.

The so-called ‘Euler angle decompositions’ were originally a way to describe rotations of the sphere in 3D, in terms of rotations about the x -, y - or z -axes. In the case of single-qubit unitary transformations, we can fairly easily describe them as a way to realise a unitary transformation $U : \mathbb{C}^2 \rightarrow \mathbb{C}^2$, in terms of the coefficients of U as a matrix.

To do this, we must rely again on the connection between single-qubit state-vectors and angular co-ordinates on the sphere. We also rely on the fact that the columns of a unitary $U : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ will be orthogonal (because $U|0\rangle$ must be orthogonal to $U|1\rangle$). Let $|u_0\rangle = U|0\rangle$, and $|u_1\rangle = U|1\rangle$. We may write:

$$|u_0\rangle = e^{i\alpha} \left(\cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi_1} \sin\left(\frac{\theta}{2}\right) |1\rangle \right) \quad (3.41)$$

for some angles θ , ϕ_1 and α , by the decomposition of Section 2.4.4 of the Week 2 study materials. The vector $-e^{-i\phi_1} \sin\left(\frac{\theta}{2}\right) |0\rangle + \cos\left(\frac{\theta}{2}\right) |1\rangle$ is then a state-vector which is orthogonal to $|u_0\rangle$, and which is different from $|u_1\rangle$ by at most a global phase factor. If we factorise away $e^{i\alpha}$ from the coefficients of U , we may then write:

$$U = e^{i\alpha} \begin{bmatrix} \cos\left(\frac{\theta}{2}\right) & -e^{i(\phi_2-\phi_1)} \sin\left(\frac{\theta}{2}\right) \\ e^{i\phi_1} \sin\left(\frac{\theta}{2}\right) & e^{i\phi_2} \cos\left(\frac{\theta}{2}\right) \end{bmatrix}, \quad (3.42)$$

for some appropriately chosen phase factor $e^{i\phi_2}$. If, as in the preceding section, we write $Q = e^{i\theta/2} R_z(\phi_1) R_y(\theta)$, we then have:

$$Q^\dagger U = e^{i\alpha} \begin{bmatrix} 1 & 0 \\ 0 & e^{i\phi_2} \end{bmatrix} = e^{i\alpha} R_z(\phi_2). \quad (3.43)$$

From this, it follows that:

$$U = e^{i\alpha} Q R_z(\phi_2) = e^{i(\alpha+\theta/2)} R_z(\phi_1) R_y(\theta) R_z(\phi_2). \quad (3.44)$$

Again, we can describe variations of the Euler decomposition involving any pair of the axes x , y and z , and in any order. For instance, to decompose U into a y -rotation between two x -rotations, we may consider a decomposition as the preceding for the unitary $HUH^\dagger = e^{i\alpha'} R_z(\phi'_2) R_y(\theta') R_z(\phi'_1)$ for some angles θ' , ϕ'_1 , ϕ'_2 and α' . It follows that:

$$\begin{aligned} U &= e^{i\alpha'} H^\dagger R_z(\phi'_2) R_y(\theta') R_z(\phi'_1) H \\ &= e^{i\alpha'} \left[H^\dagger R_z(\phi'_2) H \right] \left[H^\dagger R_y(\theta') H \right] \left[H^\dagger R_z(\phi'_1) H \right] \\ &= e^{i\alpha'} R_x(\phi'_2) R_y(-\theta') R_x(\phi'_1). \end{aligned} \quad (3.45)$$

This uses the fact that H interchanges the x - and z -axes, and also maps the positive y -axis onto the negative y -axis. We could then decompose U instead into a z -rotation between two x -rotations, by decomposing $R_y(-\theta') = R_x(-\frac{\pi}{2}) R_z(-\theta') R_x(\frac{\pi}{2})$, and simplifying the result by combining x -rotations together.

In this way, we can obtain a range of different ways to decompose a unitary transformation as a product of three ‘simpler’ unitaries, involving the x -, y - and z -axes. There are more ways beyond these to decompose a unitary operation, but the ways we have described above will suffice for us in this module.

3.2 Multi-qubit states and unitary transformations

To perform a meaningful quantum computation, it is necessary to use more than one qubit. We have already seen all of the mathematics required to describe multi-qubit states and the way that we can transform them. We're now in a position to assemble these mathematical tools to describe multi-qubit state-vectors.

In the following, we will often refer to collections of individual qubits. However, as with bit registers in RAAL, we may consider groups of qubits which we take together as a single logical piece of data. We refer to these as **quantum registers**.

3.2.1 Multi-qubit states in general

We begin with as general a description of a multi-qubit state-vector as possible, describing aspects of this definition that follows.

A state-vector on n qubits is a unit vector $|\psi\rangle \in \mathbb{C}^{2^n}$: that is, a complex-valued vector of the form:

$$\begin{aligned} |\psi\rangle &= a_{00\dots 00} |00\dots 00\rangle + a_{00\dots 01} |00\dots 01\rangle + \dots + a_{11\dots 10} |11\dots 10\rangle + a_{11\dots 11} |11\dots 11\rangle \\ &= \sum_{x \in \{0,1\}^n} a_x |x\rangle, \quad \text{where } \sum_x a_x^* a_x = \sum_x |a_x|^2 = 1. \end{aligned} \quad (3.46)$$

The reason for this is actually fairly straightforward, assuming that you are comfortable with what we described in Week 2 for single-qubit states and in Week 1 for probability distributions over bitstrings. A quantum state-vector on n qubits, may have a non-zero component $a_x |x\rangle$ for any or all of the Boolean strings $\{0,1\}^n$, and is described by the sum of all of those components. The reason why we require it to be a unit vector, is because we require $|a_x|^2$ to add up to 1. The reason why we require **that**, is because $|a_x|^2$ will correspond to the probability of obtaining x if we were to measure all of the qubits in the computational basis of $|0\rangle, |1\rangle$.

3.2.2 Product states

Just as random bits can have values which are either independent or correlated (as we saw in the Week 1 Study materials), we may consider qubits which are in independent states, or which must be described through a joint quantum state. The mathematics which we use in both the cases of random bits and quantum bits are effectively the same, though the consequences are somewhat different.

Suppose we have two qubits, the first in some state $|\alpha\rangle = a_0 |0\rangle + a_1 |1\rangle$, and the second in a state $|\beta\rangle = b_0 |0\rangle + b_1 |1\rangle$. The **product state** of the two systems, is the state of the two qubits when we consider them as a single, larger system:

$$\begin{aligned} |\psi\rangle &= |\alpha\rangle \otimes |\beta\rangle = (a_0 |0\rangle + a_1 |1\rangle) \otimes (b_0 |0\rangle + b_1 |1\rangle) \\ &= a_0 b_0 |00\rangle + a_0 b_1 |01\rangle + a_1 b_0 |10\rangle + a_1 b_1 |11\rangle. \end{aligned} \quad (3.47)$$

In the final line of the preceding equation, we essentially just expand the tensor product of the two expressions in parentheses, the way we would expand a product of the form $(a_0 + a_1)(b_0 + b_1)$.

While the coefficients a_0, a_1 and b_0, b_1 are not probabilities in this case, we compute these coefficients similarly to if they were independent probabilities. For $x, y \in \{0, 1\}$, the coefficient associated with $|x, y\rangle$ is $a_x b_y$, which is just the product of the coefficients coming from the states $|\alpha\rangle, |\beta\rangle \in \mathbb{C}^2$. (Of course, the probability $|a_x b_y|^2$ will actually be a product of the two independent probabilities $|a_x|^2$ and $|b_y|^2$, but these arise from the more-fundamental properties of the amplitudes $a_x b_y$).

We can generalise these remarks to multiple qubits: the state-vector $|\alpha\rangle \otimes |\beta\rangle \otimes |\gamma\rangle$ describes three qubits, where each one is independent of the other two. Similar remarks apply for tensor products of any number of qubits. Note that the computational basis states $|x_1 x_2 \cdots x_n\rangle$, for Boolean strings $x_1 x_2 \cdots x_n \in \{0, 1\}^n$, are product states:

$$|x_1 x_2 \cdots x_n\rangle = |x_1, x_2, \dots, x_n\rangle = |x_1\rangle \otimes |x_2\rangle \otimes \cdots \otimes |x_n\rangle. \quad (3.48)$$

We may compute tensor products of more than two qubits by grouping together the factors in any way that we like. For instance, to compute $|\alpha\rangle \otimes |\beta\rangle \otimes |\gamma\rangle$, we may compute either of $(|\alpha\rangle \otimes |\beta\rangle) \otimes |\gamma\rangle$ or $|\alpha\rangle \otimes (|\beta\rangle \otimes |\gamma\rangle)$ – the order doesn't matter. We may sometimes abbreviate states such as $|\alpha\rangle \otimes |\beta\rangle \otimes |\gamma\rangle$ by $|\alpha, \beta, \gamma\rangle$, especially when one or more of $|\alpha\rangle, |\beta\rangle, |\gamma\rangle$ are computational basis states.

Not all multi-qubit states can be described as tensor products of single-qubit states. We allow state-vectors $|\psi\rangle$ consisting of all linear combinations of computational basis states, so long as they are unit-vectors. Not all of these can be described as tensor products of other states. For example, it is not difficult to show that the following state cannot be described as a product state $|\alpha\rangle \otimes |\beta\rangle$ for any states $|\alpha\rangle = a_0 |0\rangle + a_1 |1\rangle$ or $|\beta\rangle = b_0 |0\rangle + b_1 |1\rangle$:

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle \quad (3.49)$$

However, it is still very useful to describe operations in terms of product states (and the standard basis in particular). For this reason, product states represent an important class of multi-qubit state. We consider what it means for a state **not** to be a product state, in more detail in Section 3.3.3.

3.2.3 Single-qubit operations on multi-qubit states

We may describe how to perform operations on one qubit, out of many, in the same way that we can describe how individual instructions affect probability vectors, which determine the probabilities of outcomes in RAALcr procedures.

If we have an n -qubit state $|\psi\rangle$, we can describe performing a unitary transformation U

on the k^{th} qubit by using the tensor product:

$$U_k = \underbrace{I \otimes I \otimes \cdots \otimes I}_{k-1 \text{ times}} \otimes \underbrace{U}_{k^{\text{th}} \text{ tensor factor}} \otimes \underbrace{I \otimes \cdots \otimes I}_{n-k \text{ times}} = I_{2^{k-1}} \otimes U \otimes I_{2^{n-k}}. \quad (3.50)$$

(If $k = 1$, we may omit the identity operations before the U ; and if $k = n$, we may omit the identity operations after the U . This is because $I_{2^0} = I_1 = [1]$, the 1×1 identity matrix; and $M \otimes [1] = [1] \otimes M = M$ for any matrix M .)

Remark on notation: On occasion, it can be helpful to write an operator such as $I \otimes I \otimes I \otimes U \otimes I \otimes I$, to be as clear as possible about how an operator acts on only one qubit out of many. But it is often preferable to write U_k (for some $1 \leq k \leq n$) when describing a matrix multiplication of U with a state-vector. This is in part because it saves a lot of time and space, and we will frequently do so.

To describe the effect of an operation U_k on an n -qubit state $|\psi\rangle$, we can consider the expansion of $|\psi\rangle$ into its components in the computational basis, as in Equation (3.46):

$$|\psi\rangle = a |x_1 x_2 \cdots x_n\rangle + b |y_1 y_2 \cdots y_n\rangle + c |z_1 z_2 \cdots z_n\rangle + \cdots. \quad (3.51)$$

We call this a **superposition** of the states $|x_1 x_2 \cdots x_n\rangle$, $|y_1 y_2 \cdots y_n\rangle$, $|z_1 z_2 \cdots z_n\rangle$, and so forth. (The term ‘superposition’ is flexible and is not special to multi-qubit states or the computational basis). We may refer to single-qubit states $|\psi\rangle = a|0\rangle + b|1\rangle$ as a superposition of $|0\rangle$ and $|1\rangle$ when a and b are both non-zero. Additionally, we can describe the same single-qubit state $|\psi\rangle = \left(\frac{a+b}{\sqrt{2}}\right)|+\rangle + \left(\frac{a-b}{\sqrt{2}}\right)|-\rangle$ as a superposition of $|+\rangle$ and $|-\rangle$ when $a \neq \pm b$.) When we perform an operation such as U_k on a superposition, we may evaluate the result by distributing the effect of U_k across all of the terms in the superposition:

$$U_k |\psi\rangle = a \left(U_k |x_1 x_2 \cdots x_n\rangle \right) + b \left(U_k |y_1 y_2 \cdots y_n\rangle \right) + c \left(U_k |z_1 z_2 \cdots z_n\rangle \right) + \cdots. \quad (3.52)$$

Recall that each of the vectors $|x_1 x_2 \cdots x_n\rangle$, $|y_1 y_2 \cdots y_n\rangle$, and so forth, can also be described as tensor products of the basis vectors $|0\rangle, |1\rangle \in \mathbb{C}^2$. We can then consider the transformation U_k on $|x_1 x_2 \cdots x_n\rangle$, by computing $U |x_k\rangle$ and then taking the tensor product with the other ‘tensor factors’ of $|x\rangle$:

$$U_k |x_1 x_2 \cdots x_n\rangle = |x_1\rangle \otimes |x_2\rangle \otimes \cdots \otimes |x_{k-1}\rangle \otimes \left(U |x_k\rangle \right) \otimes |x_{k+1}\rangle \otimes \cdots \otimes |x_n\rangle. \quad (3.53)$$

If $U |x_k\rangle = a|0\rangle + b|1\rangle$ has more than one term (if it is a superposition of $|0\rangle$ and $|1\rangle$), then when we take the tensor product with the other tensor factors $|x_1\rangle$, $|x_2\rangle$, and so

forth, we distribute the tensor product over the sum:

$$\begin{aligned}
U_k |x_1 x_2 \cdots x_n\rangle &= |x_1\rangle \otimes \cdots \otimes |x_{k-1}\rangle \otimes (a|0\rangle + b|1\rangle) \otimes |x_{k+1}\rangle \otimes \cdots \otimes |x_n\rangle \\
&= a |x_1\rangle \otimes \cdots \otimes |x_{k-1}\rangle \otimes |0\rangle \otimes |x_{k+1}\rangle \otimes \cdots \otimes |x_n\rangle \\
&\quad + b |x_1\rangle \otimes \cdots \otimes |x_{k-1}\rangle \otimes |1\rangle \otimes |x_{k+1}\rangle \otimes \cdots \otimes |x_n\rangle \\
&= a |x_1 x_2 \cdots x_{k-1} 0 x_{k+1} \cdots x_n\rangle + b |x_1 x_2 \cdots x_{k-1} 1 x_{k+1} \cdots x_n\rangle . \quad (3.54)
\end{aligned}$$

We can then compute $U_k |\psi\rangle$, by performing this calculation for each term in the superposition which describes $|\psi\rangle$.

Example: Consider three qubits, initially in the state $|000\rangle$. We describe the result of performing a Hadamard operation on the second qubit out of three as follows:

$$\begin{aligned}
H_2 |000\rangle &= (I \otimes H \otimes I) |000\rangle = |0\rangle \otimes (H|0\rangle) \otimes |0\rangle \\
&= |0\rangle \otimes \left(\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right) \otimes |0\rangle \\
&= \frac{1}{\sqrt{2}} |000\rangle + \frac{1}{\sqrt{2}} |010\rangle . \quad (3.55)
\end{aligned}$$

Notice how the state after the operation is now a superposition of the two state-vectors $|000\rangle$ and $|010\rangle$.

Example: Consider three qubits, initially in the state $|0, +, 0\rangle = |0\rangle \otimes |+\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}} |000\rangle + \frac{1}{\sqrt{2}} |010\rangle$. We describe the result of performing a Hadamard operation on the first qubit as follows:

$$\begin{aligned}
H_1 |0, +, 0\rangle &= (H \otimes I \otimes I) \left(\frac{1}{\sqrt{2}} |000\rangle + \frac{1}{\sqrt{2}} |010\rangle \right) \\
&= \frac{1}{\sqrt{2}} (H|0\rangle) \otimes |00\rangle + \frac{1}{\sqrt{2}} (H|0\rangle) \otimes |10\rangle \\
&= \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right) \otimes |00\rangle + \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right) \otimes |10\rangle \\
&= \frac{1}{2} |000\rangle + \frac{1}{2} |100\rangle + \frac{1}{2} |010\rangle + \frac{1}{2} |110\rangle . \quad (3.56)
\end{aligned}$$

3.2.4 Unitary transformations of more than one qubit

To perform meaningful quantum computations, we need to perform transformations of the states of the qubits, in which two or more qubits interact. As before, these will be by **unitary** transformations – operators $U : \mathbb{C}^{2^n} \rightarrow \mathbb{C}^{2^n}$ such that $U^\dagger U = I$ – because they map unit vectors to other unit vectors.

A unitary transformation on two qubits is described by a complex-valued matrix:

$$U = \begin{bmatrix} u_{0,0} & u_{0,1} & u_{0,2} & u_{0,3} \\ u_{1,0} & u_{1,1} & u_{1,2} & u_{1,3} \\ u_{2,0} & u_{2,1} & u_{2,2} & u_{2,3} \\ u_{3,0} & u_{3,1} & u_{3,2} & u_{3,3} \end{bmatrix} \quad (3.57)$$

whose columns are orthogonal unit vectors. On occasion, it is helpful to write out such a matrix explicitly, and we will consider several examples next. We could write a similar expression for a unitary transformation on three qubits – writing out a generic-looking 8×8 matrix – but we won't. Except for unitary transformations on small numbers of qubits, it is not often helpful in practice to write out a unitary matrix in full. Instead, we try to find ways of describing unitary transformations in ways that allow us to describe how they affect state-vectors in ket notation (for example, how they affect computational basis states or product states).

(a) The controlled- X , or ‘controlled-NOT’, operation

Perhaps the most commonly used two-qubit operation in quantum algorithms is the ‘controlled- X ’ operation which we write as CX . This is a unitary transformation which performs the same transformation on quantum states as the `RAALcr` instruction `CNOT` performs to probability distributions:

$$CX = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (3.58)$$

Practically speaking, it is more useful to describe this in terms of how it transforms computational basis states (which, again, is identical to the `CNOT` instruction):

$$|x, t\rangle \xrightarrow{CX} |x, t \oplus x\rangle. \quad (3.59)$$

For the CX operation, the order of the qubits is important. The first qubit that CX acts on (in other words, the first tensor factor) is called the **control**; the second qubit is called the **target**.

This operation is widely known simply as the ‘controlled-NOT’ operation and written as `CNOT`. In this module, however, we instead distinguish between `CNOT` (which acts on a pair of bits) and CX (which acts on a pair of qubits).

Example: Consider three qubits, initially in the state $|0, +, 0\rangle = |0\rangle \otimes |+\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}|000\rangle + \frac{1}{\sqrt{2}}|010\rangle$. We describe the result of performing a CX operation on the second and third qubits as follows:

$$\begin{aligned} CX_{2,3} |0, +, 0\rangle &= (I \otimes CX) \left(\frac{1}{\sqrt{2}} |000\rangle + \frac{1}{\sqrt{2}} |010\rangle \right) \\ &= \frac{1}{\sqrt{2}} |0\rangle \otimes (CX |00\rangle) + \frac{1}{\sqrt{2}} |0\rangle \otimes (CX |10\rangle) \\ &= \frac{1}{\sqrt{2}} |0\rangle \otimes |00\rangle + \frac{1}{\sqrt{2}} |0\rangle \otimes |11\rangle = \frac{1}{\sqrt{2}} |000\rangle + \frac{1}{\sqrt{2}} |011\rangle. \end{aligned} \quad (3.60)$$

Note that there is no convenient way to describe the matrix ‘ $CX_{j,k}$ ’, where the j^{th} qubit is the control and the k^{th} qubit is the target, when j and k are much larger than 2. (Ignoring only what its coefficients are, just the size of the matrix for ‘ $CX_{5,1}$ ’ makes it

impractical to write out even once). If we must, we can expand a matrix such as ‘ $CX_{j,k}$ ’ in terms of projectors, using the subscript notation to describe the qubits acted on:

$$CX_{j,k} = |0\rangle\langle 0|_j \otimes I_k + |1\rangle\langle 1|_j \otimes X_k, \quad (3.61)$$

without worrying whether $j < k$ or $j > k$ when writing the tensor factors. However, for an operation such as CX which has such a convenient description in terms of the computational basis as in Equation (3.59), it is easiest to just consider how the indexing $CX_{j,k}$ affects the way we apply that description.

Example: Consider three qubits, initially in the state $\frac{1}{\sqrt{2}}|000\rangle + \frac{1}{\sqrt{2}}|011\rangle$. We describe the result of performing a $CX_{3,1}$ operation (with the third qubit as the control and the first qubit as the target) as follows:

$$\begin{aligned} CX_{3,1} \left(\frac{1}{\sqrt{2}}|000\rangle + \frac{1}{\sqrt{2}}|011\rangle \right) &= \frac{1}{\sqrt{2}}CX_{3,1}|0,0,0\rangle + \frac{1}{\sqrt{2}}CX_{3,1}|0,1,1\rangle \\ &= \frac{1}{\sqrt{2}}|0 \oplus 0, 0, 0\rangle + \frac{1}{\sqrt{2}}|0 \oplus 1, 1, 1\rangle \\ &= \frac{1}{\sqrt{2}}|0, 0, 0\rangle + \frac{1}{\sqrt{2}}|1, 1, 1\rangle. \end{aligned} \quad (3.62)$$

(b) The ‘SWAP’ operation

There is an important two-qubit operation, which you may see if you read about operations in quantum computers where there are limitations on which qubits can interact with other qubits. That operation is the ‘SWAP’ operation:

$$\text{SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (3.63)$$

which just interchanges the states of two qubits, for example if they have states in the computational basis:

$$|x, y\rangle \xrightarrow{\text{SWAP}} |y, x\rangle. \quad (3.64)$$

We can elaborate on this, to describe what happens to states $|\alpha\rangle \otimes |\beta\rangle$, for any single-qubit states $|\alpha\rangle = a_0|0\rangle + a_1|1\rangle$ and $|\beta\rangle = b_0|0\rangle + b_1|1\rangle$:

$$|\alpha\rangle \otimes |\beta\rangle \xrightarrow{\text{SWAP}} |\beta\rangle \otimes |\alpha\rangle. \quad (3.65)$$

Note that we don’t require the input qubits to SWAP to have independent states. As with all other unitary transformations, it acts on each term of a superposition separately.

Example: Consider two qubits, initially in the state $\frac{1}{2}|01\rangle + \frac{\sqrt{3}}{2}|10\rangle$. We describe the result of performing a SWAP operation on these qubits as follows:

$$\begin{aligned} \text{SWAP} \left(\frac{1}{2}|01\rangle + \frac{\sqrt{3}}{2}|10\rangle \right) &= \frac{1}{2}\text{SWAP}|01\rangle + \frac{\sqrt{3}}{2}\text{SWAP}|10\rangle \\ &= \frac{1}{2}|10\rangle + \frac{\sqrt{3}}{2}|01\rangle = \frac{\sqrt{3}}{2}|01\rangle + \frac{1}{2}|10\rangle. \end{aligned} \quad (3.66)$$

(c) Coherently controlled single-qubit operations

The CX operation is actually one example of an important class of operations: **coherently controlled** operations. These are operations of the form:

$$CU = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & u_{00} & u_{01} \\ 0 & 0 & u_{10} & u_{11} \end{bmatrix}, \quad \text{where } U = \begin{bmatrix} u_{00} & u_{01} \\ u_{10} & u_{11} \end{bmatrix} \text{ is unitary.} \quad (3.67)$$

For example: the CX operation is a ‘controlled- X ’ operation, which is to say the operation CU in the special case of $U = X$. We consider other ‘coherently controlled’ operations, as follows:

The CY (or ‘controlled- Y ’ operation. We define an operation CY, which we may define by the matrix:

$$CY = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \end{bmatrix}. \quad (3.68)$$

We can also describe how it affects standard-basis state-vectors,

$$|x, y\rangle \xrightarrow{\text{CZ}} i^{(x-2xy)} |x, y \oplus x\rangle, \quad (3.69)$$

although this is a little bit awkward. It is much more meaningful to describe it in terms of the Pauli Y operator:

$$|x\rangle \otimes |\psi\rangle \xrightarrow{\text{CY}} |x\rangle \otimes Y^x |\psi\rangle, \quad (3.70)$$

so that the second qubit is acted on by Y if $x = 1$, and is left unaffected if $x = 0$ (and superpositions of these two cases are affected on a term-by-term basis). As with the CX operation, we describe the first qubit as the ‘control’, and the second qubit as the ‘target’.

Example: Consider two qubits, initially in the state $|+, 0\rangle = |+\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|10\rangle$. We describe the result of performing a CY operation on these qubits as follows:

$$\begin{aligned} \text{CY}(|+\rangle \otimes |0\rangle) &= \text{CY}\left(\frac{1}{\sqrt{2}}|0, 0\rangle + \frac{1}{\sqrt{2}}|1, 0\rangle\right) = \frac{1}{\sqrt{2}}\text{CY}|0, 0\rangle + \frac{1}{\sqrt{2}}\text{CY}|1, 0\rangle \\ &= \frac{1}{\sqrt{2}}|0, 0\rangle + \frac{i}{\sqrt{2}}|1, 1\rangle. \end{aligned} \quad (3.71)$$

The CZ (or ‘controlled- Z ’, or ‘controlled-phase’) operation

This is a unitary transformation which leaves the standard basis unchanged up to a phase factor, but does have an effect on other states. We may define it by the matrix:

$$CZ = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}, \quad (3.72)$$

or again by describing how it affects standard-basis state-vectors:

$$|x, y\rangle \xrightarrow{\text{CZ}} (-1)^{xy} |x, y\rangle, \quad (3.73)$$

with a phase factor of -1 if, and only if, $x = y = 1$. This may seem a strange operation to define, as it clearly does not affect standard-basis states: the factor of -1 does not represent any change to the state of the two qubits. However, it does have an effect on other states.

Example: Consider two qubits, initially in the state $|1, +\rangle = |1\rangle \otimes |+\rangle = \frac{1}{\sqrt{2}}|10\rangle + \frac{1}{\sqrt{2}}|11\rangle$. We describe the result of performing a CZ operation on these qubits as follows:

$$\begin{aligned} \text{CZ}(|1\rangle \otimes |+\rangle) &= \text{CZ}\left(\frac{1}{\sqrt{2}}|1, 0\rangle + \frac{1}{\sqrt{2}}|1, 1\rangle\right) = \frac{1}{\sqrt{2}}\text{CZ}|1, 0\rangle + \frac{1}{\sqrt{2}}\text{CZ}|1, 1\rangle \\ &= \frac{1}{\sqrt{2}}|1, 0\rangle - \frac{1}{\sqrt{2}}|1, 1\rangle = |1\rangle \otimes |-\rangle. \end{aligned} \quad (3.74)$$

As with the CY operation, we may describe this a little more explicitly in terms of the Pauli Z operator:

$$|x\rangle \otimes |\psi\rangle \xrightarrow{\text{CZ}} |x\rangle \otimes Z^x |\psi\rangle. \quad (3.75)$$

That is, CZ could be considered an operation which performs a Z operation on the second qubit if $x = 1$, and does nothing to it if $x = 0$ (with the effect on superpositions being given by the effect on each term individually). Note that while you could describe the first qubit acted on by CZ as a ‘control’ and the second one as a ‘target’, this operation is actually symmetrical in its two arguments, as Equation (3.73) shows.

Two-qubit ‘controlled’ operations generally

We can understand coherently controlled operations, as realising a form of conditional logic. We can understand this by considering the block-matrix structure that they have in common and describing it algebraically in terms of projectors:

$$CU = \begin{bmatrix} I & 0 \\ 0 & U \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & U \end{bmatrix} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U. \quad (3.76)$$

In particular, note the way that the two terms involve the projectors $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$, which isolates vectors (and components of vectors) which have $|0\rangle$ or $|1\rangle$ in the tensor factor for the control qubit. This describes the effect of the CU operation:

- on states of the form $|0\rangle \otimes |\psi\rangle$, the operation has no effect
- on states of the form $|1\rangle \otimes |\psi\rangle$, the operation transforms the state to $|1\rangle \otimes U|\psi\rangle$
- on superpositions of the two, the operation acts term-by-term.

This last property is why these operations are called ‘coherently’ controlled operations. These operations do not measure the first qubit to force it to a state $|0\rangle$ or $|1\rangle$ (for instance, they do not test on whether ‘the first qubit is $|1\rangle$ ’ in any operational sense).

(d) **The CCX (or ‘controlled-controlled- X ’, or ‘Toffoli’) operation**

The final ‘basic’ unitary operation which we consider for now, is another unitary analogue of an operation from **RAALcr**. This is the operation **CCX**, which performs the same transformation on quantum states as the **RAALcr** instruction **CCNOT** performs to probability distributions:

$$\text{CCX} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} I & 0 & 0 & 0 \\ 0 & I & 0 & 0 \\ 0 & 0 & I & 0 \\ 0 & 0 & 0 & X \end{bmatrix}. \quad (3.77)$$

On the left, we provide a description of **CCX** in terms of 2×2 blocks. This corresponds to the following decomposition in terms of projectors:

$$\text{CCX} = |00\rangle\langle 00| \otimes I + |01\rangle\langle 01| \otimes I + |10\rangle\langle 10| \otimes I + |11\rangle\langle 11| \otimes X. \quad (3.78)$$

In this case, it is again more useful to describe this in terms of how it transforms computational basis states (which is identical to the **CCNOT** instruction):

$$|x, y, t\rangle \xrightarrow{\text{CCX}} |x, y, t \oplus (x \& y)\rangle. \quad (3.79)$$

Here, the first **and** second qubits are described as ‘controls’ and the third qubit is described as a ‘target’. This can be understood using an alternative description of how **CCX** acts on product states:

$$|x\rangle \otimes |y\rangle \otimes |\psi\rangle \xrightarrow{\text{CCX}} |x\rangle \otimes |y\rangle \otimes X^{xy} |\psi\rangle, \quad (3.80)$$

where the exponent of X is x times y .

While we describe **CCX** above as a ‘basic’ operation, it is in fact possible to decompose it as a sequence of H , $R_{\frac{\pi}{4}}$, and **CX** operations: see section 4.3 of Reference [1]. We consider it an important enough operator, to include it (slightly redundantly) among a list of basic operations to realise more-complicated unitary transformations.

3.2.5 Reversible computation using unitary transformations

An important, early result in quantum computation was to answer an important question that people do not seem to ask very often with the current excitement over quantum computers:

Can an idealised quantum computer, even compute everything that a classical computer can compute?

Fortunately, the answer is ‘yes’. The way that we can show this, is by using the same result as for reversible computation.

We have already seen that the basic reversible instructions of **RAAL** – the operations **NOT**, **CNOT**, and **CCNOT** – have analogues which are unitary transformations on qubits: the operations **X**, **CX** and **CCX**. (In particular, the matrices describing these unitary operations are the same as the matrices which describe how **NOT**, **CNOT** and **CCNOT** act on probability vectors). This means that anything that we can compute reversibly, we can also compute using unitary transformation with the **X**, **CX**, and **CCX** operations, provided that we have enough extra qubits for workspace.

In fact, we can do a simple translation from a subroutine to compute a function in **RAAL**, to a sequence of unitary transformations and measurements. There is a little bit of extra work that a quantum procedure must do, that a classical computation does not. Specifically, this is loading the classical input into the qubits (for example, by performing **X** operations on qubits to be initialised in the state $|1\rangle$) and measuring the final state to produce the classical output. We can even simulate the **RAALcr** instruction ‘**!randomise a**’ to generate random bits, by preparing ‘**a**’ in either the $|+\rangle$ or $|-\rangle$ state, measuring it along the z -axis, and recording the result each time. The result is that, using the relationship between reversible computations and permutation operations, a sufficiently advanced quantum computer can realise any computation done with a ‘normal’ computer.

3.2.6 Universality

An important result (but one that we will not describe in detail), is that it is possible to realise **any** unitary transformation on n qubits, using just the operation **CX** and any two of the operations $R_x(\alpha)$, $R_y(\alpha)$ and $R_z(\alpha)$ for various angles α . (We can even approximate any unitary transformation as closely as we like, using **CX** and any two of $R_x(\frac{\pi}{4})$, $R_y(\frac{\pi}{4})$ and $R_z(\frac{\pi}{4})$. Any imprecision in realising the unitary transformations only accumulate slowly.). The proofs of these facts can be found in section 4.5 of Reference [1].

The operations which we consider in the preceding material are then not just operations which can be easily described and enough to realise any classical computation, but also enough to realise any quantum computation in principle. The actual challenge, of course, is then to find a way to realise a given unitary transformation **efficiently**. In other words, this would be by using as few of these basic operations as possible and without using too many additional qubits as workspace.

3.2.7 Quantum parallelism?

There are some important theoretical differences between performing a computation on a classical computer and realising the same computation as a unitary transformation on a quantum computer. One of these is well known by popular science and technology enthusiasts but is also widely misunderstood: the idea that quantum computers can coherently perform a computation on a superposition.

Because unitary transformations act linearly across superpositions, it means that – in a sense – those transformations are computing with more than one value simultaneously. In fact, an n -qubit state can involve a superposition of up to 2^n different components $|x_1x_2 \cdots x_n\rangle$. This is sometimes described as a quantum form of ‘massive parallelism’.

However, there is a catch. Even if it were somehow true that a quantum computer **actually** performs many computations in parallel, we cannot simply access all of the things being ‘computed’ in this way. The reason is effectively the same as for randomised computation. If we describe a (classical!) probability vector in ket notation, **that also** is a description in terms of up to 2^n different components $|x_1x_2 \cdots x_n\rangle$, for a computation on n bits. Additionally, the computations on them are also linear transformations, which act on these components ‘in parallel’. Unfortunately, we don’t get to choose which term is actually realised in a given repetition of that computation. Similarly, because we can only obtain information from a quantum computer by measurement, we can’t access all of the results of the computation (or even a single potentially helpful result). Actually, we get a **random** result, according to some probability distribution.

Given this, you might wonder why we would want to use quantum computers at all, if all they produce are the same results as randomised computations. The answer is that simply performing computations ‘massively in parallel’ is not what quantum computers are good for. Quantum states are richer – allowing more complexity – than probability vectors and can be transformed in ways that cannot be sensibly described as operations on bits. This allows quantum computers to take ‘shortcuts’ which are not available to classical computers. This allows them to solve problems in a different way from classical computers. Although often one of the ‘ingredients’ of these ‘shortcuts’ is the ability to perform a computation on a superposition of computational basis states.

We can perhaps begin to see how limitations on the idea of ‘quantum parallelism’, by considering how measurements work on multi-qubit states. Understanding how quantum computers can take ‘shortcuts’ is the subject of much of the rest of this module.

3.3 Measurement and multi-qubit states

To obtain an actual result from a quantum computation, it is necessary to perform a measurement on one or more qubits in the computation. We now consider how to represent this process and the way that it affects quantum states similarly to the evaluation of conditional probabilities.

3.3.1 Single-qubit measurements and state collapse

When we measure a single qubit, which has a state $|\psi\rangle$ independent of any other qubits, we may describe this as we did in the Week 2 Study materials. If $|u_0\rangle, |u_1\rangle \in \mathbb{C}^2$ of the measurement, we compute the values of $|\langle u_0 | \psi \rangle|^2$ and $|\langle u_1 | \psi \rangle|^2$, which determines the probability of either outcome; and afterwards the qubit is in either the state $|u_0\rangle$ or $|u_1\rangle$. For multi-qubit states, when we measure one of the qubits involved (and obtain some

outcome), there is the question of what state the rest of the qubits are in, **conditioned** on the outcome of the measurement.

This is mathematically similar to the situation with probability distributions, in which we consider conditional distributions of some bits depending on some given values for other bits, as described in Section 1.3.6 of the Week 1 study materials. We now describe this notion of ‘conditional state’ – more commonly described as a **state collapse** – in a little detail.

(a) A three-qubit example

We illustrate this with an example of a state $|\psi\rangle \in \mathbb{C}^8$ on three qubits. Suppose that we are interested in measuring the first qubit, in the standard basis $|0\rangle, |1\rangle$. We can describe the state $|\psi\rangle$ in terms of its components in the standard basis:

$$\begin{aligned} |\psi\rangle &= a_{000}|000\rangle + a_{001}|001\rangle + a_{010}|010\rangle + a_{011}|011\rangle \\ &\quad + a_{100}|100\rangle + a_{101}|101\rangle + a_{110}|110\rangle + a_{111}|111\rangle . \end{aligned} \quad (3.81)$$

We may consider the two vectors $\mathbf{q}_0 \in \mathbb{C}^4$ and $\mathbf{q}_1 \in \mathbb{C}^4$ which we get, by taking the coefficients from $|\psi\rangle$ which are consistent with the first qubit having a ‘value’ of either 0 or 1:

$$\begin{aligned} \mathbf{q}_0 &= \sum_{x \in \{0,1\}^2} a_{0x} |x\rangle = a_{000}|00\rangle + a_{001}|01\rangle + a_{010}|10\rangle + a_{011}|11\rangle , \\ \mathbf{q}_1 &= \sum_{x \in \{0,1\}^2} a_{1x} |x\rangle = a_{100}|00\rangle + a_{101}|01\rangle + a_{110}|10\rangle + a_{111}|11\rangle . \end{aligned} \quad (3.82)$$

Note that $\mathbf{q}_0$ and $\mathbf{q}_1$ may be subnormalised, in the sense that $\|\mathbf{q}_0\|, \|\mathbf{q}_1\| < 1$. (In fact, they will **both** be subnormalised, unless one of $\mathbf{q}_0$ or $\mathbf{q}_1$ is the zero vector). Let $c_0 = \|\mathbf{q}_0\|$ and $c_1 = \|\mathbf{q}_1\|$ and (assuming that $c_0, c_1 \neq 0$) let $|\psi_0\rangle = \frac{1}{c_0}\mathbf{q}_0$ and $|\psi_1\rangle = \frac{1}{c_1}\mathbf{q}_1$ be normalised versions of the vectors $\mathbf{q}_0$ and $\mathbf{q}_1$ obtained by rescaling. Then, we may decompose $|\psi\rangle$ by splitting its coefficients as follows:

$$\begin{aligned} |\psi\rangle &= |0\rangle \otimes (a_{000}|00\rangle + a_{001}|01\rangle + a_{010}|10\rangle + a_{011}|11\rangle) \\ &\quad + |1\rangle \otimes (a_{100}|00\rangle + a_{101}|01\rangle + a_{110}|10\rangle + a_{111}|11\rangle) \\ &= |0\rangle \otimes \mathbf{q}_0 + |1\rangle \otimes \mathbf{q}_1 = c_0 |0\rangle \otimes |\psi_0\rangle + c_1 |1\rangle \otimes |\psi_1\rangle . \end{aligned} \quad (3.83)$$

If we measure the first qubit in the computational basis $|0\rangle, |1\rangle$, the effect is for the state of all three qubits to change to one of the two terms. The state $|\psi'\rangle$ after measurement is:

$$|\psi'\rangle = \begin{cases} |0\rangle \otimes |\psi_0\rangle , & \text{if the outcome is ‘0’;} \\ |1\rangle \otimes |\psi_1\rangle , & \text{if the outcome is ‘1’} . \end{cases} \quad (3.84)$$

The probability with which the two outcomes occur is determined by the coefficients c_0 and c_1 which scale the two terms. The probability of measuring ‘0’ is $|c_0|^2$, and the probability of measuring ‘1’ is $|c_1|^2$.

(b) Comparison to conditional probabilities

Quantum state collapse is sometimes presented as something strange or amazing. But, in a sense, it is simply an analogue of a conditional probability.

As we saw in Equation (1.57) in the Week 1 Study materials, the conditional probability of a register B having some value x , conditioned on another register A having the value y , as:

$$\Pr[B=x \mid A=y] = \frac{\Pr[B=x \text{ and } A=y]}{\Pr[A=y]}. \quad (3.85)$$

Suppose A is a single bit and we're considering the distribution of B depending on whether $A = 0$ or $A = 1$. Suppose that A and B have the joint distribution $\mathbf{p}$. Rather than the probabilities of specific values of the registers, we use a projector $|0\rangle\langle 0|$ or $|1\rangle\langle 1|$ on the 'tensor factor' corresponding to the register A. If $\mathbf{p}$ is a probability vector over the values of (A,B), so that A is represented by the first bit in the indexing of the coefficients of $\mathbf{p}$, we would define:

$$\mathbf{d}_0 = (|0\rangle\langle 0| \otimes I) \mathbf{p}, \quad \mathbf{d}_1 = (|1\rangle\langle 1| \otimes I) \mathbf{p}. \quad (3.86)$$

The probability that $A = 0$ is determined by the total of the coefficients in $\mathbf{d}_0$, which contains all outcomes (and only those outcomes) where $A = 0$. We might even define a special notation for this, by writing 'Total($\mathbf{d}_0$) = $\sum_x \langle x | \mathbf{d}_0$ ' for the sum of the coefficients of $\mathbf{d}_0$. Similarly, the probability that $A = 1$ would then be written as Total($\mathbf{d}_1$). For any $y \in \{0, 1\}$, the distribution $\mathbf{p}'$ of values for (A,B), given that $A = y$ (where the probability of this happening is non-zero), would be given by:

$$\mathbf{p}' = \frac{\mathbf{d}'_y}{\text{Total}(\mathbf{d}'_y)} = \frac{(|y\rangle\langle y| \otimes I) \mathbf{p}}{\text{Total}((|y\rangle\langle y| \otimes I) \mathbf{p})}. \quad (3.87)$$

Nothing mysterious happens to the register B if we learn the value of A. At most, we have learned more information about the distribution of B as a result of learning about A.

From a physical perspective, there is likely more to the story than this when we perform a measurement on a quantum state but, mathematically, the situation is almost the same. A quantum state-vector $|\psi\rangle$ does not represent probabilities directly, so we are not interested in 'Total($(|y\rangle\langle y| \otimes I) |\psi\rangle$)', but rather the **norm** of the vector $(|y\rangle\langle y| \otimes I) |\psi\rangle$,

$$\begin{aligned} \left\| (|y\rangle\langle y| \otimes I) |\psi\rangle \right\| &= \sqrt{\left[(|y\rangle\langle y| \otimes I) |\psi\rangle \right]^\dagger \left[(|y\rangle\langle y| \otimes I) |\psi\rangle \right]} \\ &= \sqrt{\langle \psi | (|y\rangle\langle y| \otimes I) (|y\rangle\langle y| \otimes I) |\psi \rangle} = \sqrt{\langle \psi | (|y\rangle\langle y| \otimes I) |\psi \rangle}. \end{aligned} \quad (3.88)$$

Here, we have used the fact that $(|y\rangle\langle y| \otimes I) (|y\rangle\langle y| \otimes I) = (|y\rangle\langle y| |y\rangle\langle y| \otimes I) = (|y\rangle\langle y| \otimes I)$ to simplify the expression under the radical. Then, if (A,B) are in a quantum state $|\psi\rangle$,

where $\mathbf{A}$ is one qubit which we measure in the computational basis, then the state of $(\mathbf{A}, \mathbf{B})$ after measuring the outcome $y \in \{0, 1\}$ is:

$$|\psi'\rangle = \frac{(|y\rangle\langle y| \otimes I) |\psi\rangle}{\|(|y\rangle\langle y| \otimes I) |\psi\rangle\|} = \frac{(|y\rangle\langle y| \otimes I) |\psi\rangle}{\sqrt{\langle\psi| (|y\rangle\langle y| \otimes I) |\psi\rangle}}, \quad (3.89)$$

provided that the probability $\|(|y\rangle\langle y| \otimes I) |\psi\rangle\|^2 = \langle\psi| (|y\rangle\langle y| \otimes I) |\psi\rangle$ of obtaining the outcome y is not zero.

3.3.2 Projective measurements in general and the Born rule

We may generalise this notion to a description of what happens when we measure a single qubit out of many – where the qubit does not have to be the first in a sequence of qubits and where that measurement does not have to be in the standard basis. We can even generalise this to measurements of systems more complicated than a single qubit.

Suppose that we want to measure some qubit $\mathbf{x}$, which is the k^{th} qubit in a sequence of n qubits, $\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_k, \dots, \mathbf{q}_n$. Let $|u_0\rangle, |u_1\rangle$ be the basis of measurement for $\mathbf{x}$. Then we may define projectors:

$$\Pi_0 = I_{2^{k-1}} \otimes |u_0\rangle\langle u_0| \otimes I_{2^{n-k}}, \quad \Pi_1 = I_{2^{k-1}} \otimes |u_1\rangle\langle u_1| \otimes I_{2^{n-k}}. \quad (3.90)$$

(If $k - 1 = 0$, the tensor factor $I_{2^{k-1}}$ can be ignored, so that $\Pi_a = |u_a\rangle\langle u_a| \otimes I_{2^{n-1}}$ and similarly if $n - k = 0$, we may write $\Pi_a = I_{2^{n-1}} \otimes |u_a\rangle\langle u_a|$). If we have a system in an n -qubit state $|\psi\rangle$, Similarly, we perform a $|u_0\rangle, |u_1\rangle$ measurement on $\mathbf{x}$, then we obtain the outcome $a \in \{0, 1\}$ with probability $\|\Pi_a |\psi\rangle\|^2 = \langle\psi| \Pi_a |\psi\rangle$, in which case the state after measurement will be:

$$|\psi'\rangle = \frac{\Pi_a |\psi\rangle}{\|\Pi_a |\psi\rangle\|} = \frac{\Pi_a |\psi\rangle}{\sqrt{\langle\psi| \Pi_a |\psi\rangle}}. \quad (3.91)$$

Notice that no matter what the probability is of obtaining the outcome $a \in \{0, 1\}$ (so long as it is non-zero), this formula ensures that the post-measurement state **is** a state – that is to say, a unit vector). It then describes a condition that the system is in after measurement, which could produce a probability distribution of outcomes if we were to measure it again.

We can generalise this even further. The formula in Equation (3.91) and the probability $\|\Pi_a |\psi\rangle\|^2 = \langle\psi| \Pi_a |\psi\rangle$, is still well defined for any projector Π_a whatsoever.

(a) ‘Total’ measurement in the computational basis

We can describe measuring n qubits in the computational basis, all at once. (We can also measure them one at a time to obtain the same result. However, in some cases, it is convenient to shortcut this description to consider the result as the outcome of a single

measurement). This is sometimes described as a **total** measurement, in the computational basis.

To describe this, we consider the possible outcomes $x \in \{0, 1\}^n$ and the corresponding projectors $\Pi_x = |x\rangle\langle x| = |x_1 x_2 \cdots x_n\rangle\langle x_1 x_2 \cdots x_n|$ on n -qubit state-vectors. The probability of obtaining the outcome $x \in \{0, 1\}^n$ when measuring n qubits in the state $|\psi\rangle$, is then precisely $\langle\psi|\Pi_x|\psi\rangle$. If $|\psi\rangle$ is an n -qubit state, this is then just the same as $\langle\psi|x\rangle\langle x|\psi\rangle = |\langle x|\psi\rangle|^2$.

If $|\psi\rangle$ is actually a state-vector on more than n qubits, there are qubits which are not directly involved in the measurement. Then the state of the whole system is projected onto those components of $|\psi\rangle$ which are consistent with the measurement outcome x on the measured qubits – renormalising if necessary to obtain a unit vector $|\psi'\rangle$ describing the post-measurement state.

(b) Measurement of an ‘observable’, and the Born rule

As we mentioned in Section 2.3.2 of the Week 2 Study materials, any ‘measurable’ property is represented by a Hermitian matrix M , and the outcome of such a measurement is some eigenvalue $a \in \mathbb{R}$ of M . In this context, we call M an **observable**.

We may consider the case where Π_a is the projector onto the set of vectors which are eigenvalues of M with that eigenvalue a . (If M has eigenvectors $|u_k\rangle, |u_{k'}\rangle, |u_{k''}\rangle, \dots$, which all have eigenvalue a , we then have $\Pi_a = |u_k\rangle\langle u_k| + |u_{k'}\rangle\langle u_{k'}| + |u_{k''}\rangle\langle u_{k''}| + \dots$). We can describe a ‘measurement of M ’, as producing an outcome which is some eigenvalue a of M , projecting an input state $|\psi\rangle$ onto the ‘ a -eigenspace’ in doing so.

When measuring an observable M , the probability of obtaining a specific outcome a is once again $\langle\psi|\Pi_a|\psi\rangle$. If this is non-zero, the post-measurement state conditioned on the outcome a is given by:

$$|\psi'\rangle = \frac{\Pi_a |\psi\rangle}{\|\Pi_a |\psi\rangle\|} = \frac{\Pi_a |\psi\rangle}{\sqrt{\langle\psi|\Pi_a|\psi\rangle}}, \quad (3.92)$$

which generalises Equation (3.91) to projectors beyond ones acting on single qubits.

This general description of measurements in terms of projections is called the **Born rule**. We will most often consider the case of measurement of a single qubit in some basis $|u_0\rangle, |u_1\rangle$, or measurement of several qubits in the computational basis. However, the general description of a measurement is important to understand the more-interesting features of quantum information theory (some of which we might even touch on in this module).

3.3.3 ‘Entanglement’, and measurement

We have seen how to describe product states, and have described something of what it means for a state to be a product state. But what would it mean, for two or more qubits

not to be independent?

For a two-qubit state $|\psi\rangle = u_{00}|00\rangle + u_{01}|01\rangle + u_{10}|10\rangle + u_{11}|11\rangle$, not being a product state simply means that the amplitude associated with the pair of bit-values x, y can't be described by $a_x b_y$ for some set of coefficients $\{a_0, a_1\}$ describing just the first qubit and some set of coefficients $\{b_0, b_1\}$ describing the second bit. On a deeper intuitive level, it means that the first and second qubits do not **have** states that can be adequately described without the other. For this reason, a pair of qubits (or any quantum system) in a state which is not a product state, is said to be **entangled**.

- For a pair of entangled qubits, there **is** a way to mathematically represent the condition of each qubit individually, as some probability distribution **in addition to** the quantum description using the Bloch sphere. But this representation is no longer by a state of the system which can be distinguished from a single other state.
- At the same time, a pair of entangled qubits **is** in some definite state, that does not involve any randomness at all. Specifically, through a well-chosen unitary transformation, it is possible to perfectly distinguish that state from multiple other states, and in this sense it is not inherently random at all.

Practically, what it means for two qubits (or any two quantum systems) to be entangled, is that measurements on them may give rise to correlated results. If we measure the two qubits (possibly using different choices of measurement for the two qubits), the outcome of the measurement on the first qubit may not be independent of the outcome of the measurement on the second qubit.

Measurement of a product state

For a system of n qubits, in a state $|\psi\rangle = |\alpha\rangle \otimes |\beta\rangle$ for some k -qubit state $|\alpha\rangle$ and $(n-k)$ -qubit state $|\beta\rangle$, a measurement on the first k qubits has no effect on the remaining $n-k$ qubits. For any k -qubit projector Π_a representing some measurement outcome a , the probability with which that outcome occurs is:

$$p_a = \langle\psi| (\Pi_a \otimes I) |\psi\rangle = \langle\alpha| \Pi_a |\alpha\rangle \otimes \langle\beta| \beta\rangle = \langle\alpha| \Pi_a |\alpha\rangle. \quad (3.93)$$

So long as $p_a \neq 0$, the post-measurement state is then:

$$|\psi'\rangle = \frac{(\Pi_a \otimes I) |\psi\rangle}{\sqrt{\langle\psi| (\Pi_a \otimes I) |\psi\rangle}} = \frac{(\Pi_a |\alpha\rangle) \otimes |\beta\rangle}{\sqrt{\langle\alpha| \Pi_a |\alpha\rangle}} = \frac{1}{\sqrt{p_a}} \Pi_a |\alpha\rangle \otimes |\beta\rangle \quad (3.94)$$

That is, the final $n-k$ qubits remain in the state $|\beta\rangle$ after measurement. If we measure those final $n-k$ qubits with some other measurement, the outcome will have a probability distribution which does not depend on a .

Measurement of an entangled state For an entangled state $|\psi\rangle$, the measurement outcomes will not generally be independent. For instance, if $a \in \{0, 1\}^k$ is a binary string and $\Pi_a = |a\rangle\langle a| = |a_1 a_2 \cdots a_k\rangle\langle a_1 a_2 \cdots a_k|$, we may describe $|\psi\rangle$ in the form:

$$|\psi\rangle = \sum_{a \in \{0,1\}^k} c_a |a\rangle \otimes |\psi_a\rangle, \quad \text{where } c_a = \sqrt{\langle\psi| (|a\rangle\langle a| \otimes I) |\psi\rangle}. \quad (3.95)$$

Then the probability with which we obtain the outcome a on the first k qubits, is $c_a^2 = \langle \psi | (|a\rangle\langle a| \otimes I) | \psi \rangle$ and the post-measurement state on the final $n - k$ qubits is $|\psi_a\rangle$. Depending on how much the states $|\psi_a\rangle$ differ, measurements on the remaining $n - k$ qubits may yield results that depend significantly on the outcome a . Additionally, if we perform a different measurement to begin with on the first k qubits, we can obtain completely different states of the remaining $n - k$ qubits.

Example: Consider the state $|G\rangle = \frac{1}{\sqrt{2}} |000\rangle + \frac{1}{\sqrt{2}} |111\rangle$.

- If we measure the first qubit in the computational basis, we obtain either the outcome ‘0’ or ‘1’, each with equal probability $p = 1/2$. If the outcome is ‘0’, the state we obtain post-measurement is:

$$|G'\rangle = \frac{(|0\rangle\langle 0| \otimes I) |G\rangle}{\sqrt{\langle G | (|0\rangle\langle 0| \otimes I) |G\rangle}} = |000\rangle. \quad (3.96)$$

Similarly, if the outcome is ‘1’, the post-measurement state is $|111\rangle$. Then, after measurement, the second and third qubits have states identical to the first qubit.

- If we measure the first qubit in the $|+\rangle, |-\rangle$ basis, it is not difficult to show that again the two outcomes occur with equal probability, for example:

$$\begin{aligned} & \langle G | (|+\rangle\langle +| \otimes I) |G\rangle \\ &= \frac{1}{2} \left(\langle 0 | + \rangle \langle + | 0 \rangle \otimes \langle 00 | 00 \rangle + \langle 0 | + \rangle \langle + | 1 \rangle \otimes \langle 00 | 11 \rangle \right. \\ & \quad \left. + \langle 1 | + \rangle \langle + | 0 \rangle \otimes \langle 11 | 00 \rangle + \langle 1 | + \rangle \langle + | 1 \rangle \otimes \langle 11 | 11 \rangle \right) \\ &= \frac{1}{2} \left(\langle 0 | + \rangle \langle + | 0 \rangle + \langle 1 | + \rangle \langle + | 1 \rangle \right) = \frac{1}{2} \cdot \left(\frac{1}{2} + \frac{1}{2} \right) = \frac{1}{2}; \end{aligned} \quad (3.97)$$

and similarly for the outcome $|-\rangle$. For the outcome $|+\rangle$, the state after measurement is then given by:

$$\begin{aligned} |G'\rangle &= \frac{(|+\rangle\langle +| \otimes I) |G\rangle}{\sqrt{\langle G | (|+\rangle\langle +| \otimes I) |G\rangle}} \\ &= \sqrt{2} \cdot \left[\frac{1}{\sqrt{2}} (|+\rangle\langle +| \otimes I) |000\rangle + \frac{1}{\sqrt{2}} (|+\rangle\langle +| \otimes I) |111\rangle \right] \\ &= \sqrt{2} \cdot \left[\frac{1}{2} |+\rangle \otimes |00\rangle + \frac{1}{2} |+\rangle \otimes |11\rangle \right] = |+\rangle \otimes \left[\frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle \right]; \end{aligned} \quad (3.98)$$

a similar calculation would show that for the outcome $|-\rangle$, the post-measurement state would be:

$$|G''\rangle = \frac{(|-\rangle\langle -| \otimes I) |G\rangle}{\sqrt{\langle G | (|-\rangle\langle -| \otimes I) |G\rangle}} = |-\rangle \otimes \left[\frac{1}{\sqrt{2}} |00\rangle - \frac{1}{\sqrt{2}} |11\rangle \right]. \quad (3.99)$$

The relationship between entangled states and measurement is important for quantum information theory in general. In particular, it is possible to show that entanglement allows for forms of correlation between two qubits, which cannot be directly simulated using classical randomness without extra communication. Exactly what quantum states represent has been the cause of much debate among physicists. However, as we are acting in the capacity of computer scientists here, we will consider this fact to simply be a curiosity with potentially interesting applications. It will inform the way that we write quantum algorithms, to make sure that we get the most useful possible outcome from our computation.

Summary

Well done! You have now reached the end of this week's Study – Week 3: Single-qubit unitary transformations and multi-qubit states and unitaries.

We have advanced what we learned last week on single-qubit states and measurements, and have covered measurement and unitaries of multi-qubit states. Next week, we will learn about many-qubit procedures and look at examples and intuitions.

References

[1] Nielsen, M. A. and Chuang, I. L. Quantum Computation and Quantum Information. 10th anniversary edn. Cambridge, UK: Cambridge University Press, 2010, p. 159.